| Qn | Suggested Answer |
|---|---|
| 1 | |
| (a) | *Precision* is defined as a measure of how close the experimental values are to each other. |
| | *Accuracy* is defined as a measure of how close the experimental values are to the true value of the physical quantity. |
| (b) | $g = \dfrac{4\pi^2 L}{T^2} = \dfrac{4\pi^2 \left(50.0 \times 10^{-2}\right)}{1.42^2} = 9.78933 \text{ m s}^{-2}$ |
| | $\pm\dfrac{\Delta g}{g} = \pm\left(\dfrac{\Delta L}{L} + 2\dfrac{\Delta T}{T}\right)$<br>$\pm\Delta g = \pm\left(\dfrac{0.2}{50.0} + \dfrac{0.02}{1.42}\right) \times 9.78933 = \pm 0.17704$ |
| | $= \pm\ 0.2 \text{ m s}^{-2}$ (1 s.f.)<br>$g = 9.8 \pm 0.2 \text{ m s}^{-2}$ |
| (c) | There is error due to human reaction time. Taking a large number of oscillations will reduce the fractional/percentage uncertainty of the measurement of time. The absolute uncertainty is the same but taking more oscillations reduces the effect in the calculation of $T$.<br>($\Delta T = \Delta t\ /\ n$ where $\Delta t = \pm\ 0.2 – 0.4$ s (human reaction time) and $n$ is the number of oscillations.) |
| | |
| 2 | |